

D-He³ Fuel Cycle for a Space Propulsion System

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An open-ended magnetic confinement fusion system which appears to lend itself nicely to a space propulsion system that can meet the stringent requirements of space exploration is examined. The system is based on a magnetic mirror configuration which has been studied for decades in connection with applications to terrestrial power reactors and whose confinement physics is reasonably well understood. What distinguishes this propulsion device from the conventional mirror reactor is the fact that the plasma in it is sufficiently dense as to make the collision mean free path much smaller than the dimension of the system, thereby allowing the plasma to behave much like a fluid. Although D-T fuel cycle is desirable in early stages of development because of its low ignition temperature and less stringent technological requirements, D-He³ is most often mentioned due to the fact that its reaction products are charged particles and its neutron production is minimal. The plasma dynamics of a D-He³ mirror system can be obtained from a set of conservation equations which yields, among other things, a reactor gain factor Q greater than unity with a very high injection energy ≥ 200 keV. When applied to the propulsion system considered in this work, it is seen that D-He³ fuel produces a specific impulse exceeding 4×10^5 s and a thrust on the order of 10^6 N.

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I. INTRODUCTION

Much interest in human interplanetary exploration has been generated as numerous probes, orbitors, and landers have been launched and performed their missions successfully in the recent years [1–5]. Because crewed space travel is hazardous due to galactic radiation, emphasis is predominantly placed on propulsion systems that allow the mission to be undertaken in a relatively short time to minimize the risk. Neither conventional propulsion systems nor some of the nuclear thermal systems can meet this requirement due to the small specific impulse they generate. With antimatter annihilation devices still far in the future, nuclear-fusion-based systems appear to be the likely candidates because they have the sufficiently large energy content per unit mass to provide the propulsive characteristics needed for fast travel times. Fusion has been studied for several decades for terrestrial power applications [6–8]. This traditional fusion research effort has provided the basis for the potential propulsion systems. These fall under two major categories: magnetic confinement [9–11] and inertial confinement [12, 13] approaches. Recently, many concepts utilizing combined features of both the inertial and the magnetic fusion approaches, such as magnetically in-

sulated inertial confinement fusion (MICF) [14, 15] and magnetized target fusion (MTF) [16, 17], have been proposed. Antimatter-driven fusion propulsion is also being actively studied by many scientists [18, 19].

In this paper we examine a device based on magnetic confinement, and it consists of a magnetic mirror configuration where the magnetic field strength at the end is much larger than that at the center to provide the needed plasma containment. More specifically, it is a device which confines a hot plasma long enough to generate fusion energy while allowing a certain fraction of its charged-particle population to escape from one end to produce thrust. Under these conditions, the plasma behaves like a continuous medium - a fluid, and its confinement is governed by gas dynamic laws. The escape of the plasma from the system is analogous to the flow of a gas into a vacuum from a vessel with a hole. This type of device, named the gasdynamic mirror (GDM), was first proposed and studied by Kammash *et al.* for space propulsion applications [20–22]. Accordingly, for such a GDM system, the particle lifetime or plasma confinement time can be expressed as

$$\tau = \frac{RL}{v_{th}}, \quad (1)$$

where R is the mirror ratio seen by the plasma, L the plasma length (mirror length), and v_{th} the thermal velocity. With the aid of a power flow diagram for a propul-

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sion system that consists of a reactor, an injector, and a thermal and a direct converter, it has been shown that the energy multiplication factor of the fusion reactor (Q value) has to be slightly larger than unity to accommodate the efficiencies of the various components [21]. A preliminary analysis of such a propulsion system reveals that for a deuterium-tritium (D-T) fuel cycle, the reactor will be about 44 m long with a total vehicle mass of 422 metric tons [20]. It was shown in Ref. 20 that with a D-T fueled GDM, a round trip to Mars could be done in around 170 days with a specific impulse of 1.1×10^5 seconds and a thrust of 5.0×10^4 N.

Although the GDM with D-T fuel has been well demonstrated, the D-T reaction produces high-energy neutrons which require additional shielding to protect system components, placing a significant mass burden on the vehicle. Moreover, tritium is a radioactive fuel which creates safety concerns during launch into orbit or possible re-entry into the Earth's atmosphere. This paper will focus on the advanced fuel cycle D-He³. Although the D-He³ reaction still produces some neutrons, the number will be significantly reduced compared with the D-T option. Although D-He³ will require a higher ignition temperature than the D-T option, it is still achievable, for example, with the D-T trigger. Once initiated, however, the D-He³ fusion reaction offers both an improved level of energy release, compared to the D-D baseline, and a lower neutron flux. With both an improved energy release and a larger proportion of liberated energy appearing in the form of charged-particle kinetic energy, significant performance improvements will result. The governing equations of the GDM are the particle and the energy conservation equations, which when solved self-consistently yield the length of the device as a function of the plasma parameters.

II. ADVANCED FUEL CYCLE

The advanced fuel cycle studied in this paper is the reaction of deuterium and helium-3 (D-He³):



Whereas the neutron power constitutes about 80 % of the fusion power in a typical D-T reaction, D-He³ makes up about 5 % or less in D-He³ reactions, and such a reaction should alleviate considerably problems associated with disposal of this heat component. The D-He³ reaction itself does not produce neutrons, but satellite reactions involving D do, indeed, produce neutrons although to a lesser extent. These favorable aspects of D-He³ reactions are, however, offset by the high ignition temperature required by this fuel cycle and by the corresponding large amounts of synchrotron radiation emitted.

For the analysis, the particle and the energy conservation equations are adopted and solved self-consistently to yield some plasma characteristics. If we denote the

fuel ion density by n_i and define the concentrations of D and He³ as C_D and C_{He^3} , respectively, so that

$$C_D = n_D/n_i \quad \text{and} \quad C_{He^3} = n_{He^3}/n_i, \quad (3)$$

where

$$C_D + C_{He^3} = 1, \quad (4)$$

then the mass and the energy conservation equations for this species can be written as

$$\frac{dn_i}{dt} = S_i - \frac{n_i}{\tau_i} - 2C_D C_{He^3} n_i^2 \langle \sigma v \rangle_{DHe^3} \quad (5)$$

and

$$\begin{aligned} \frac{d(n_i E_i)}{dt} = & S_i E_{in} - \frac{n_i}{\tau_i} E_{Li} - 2C_D C_{He^3} n_i^2 \langle \sigma v \rangle_{DHe^3} E_i \\ & + W_{f\alpha i} + W_{fp i} + W_{ei} + W_{\alpha i} + W_{pi}, \end{aligned} \quad (6)$$

where S_i is the injection rate of fuel ions per unit volume, τ_i the lifetime or confinement time of the ions, E_i the ion energy, E_{in} the ion injection energy, E_{Li} the ion energy with which they escape confinement, $\langle \sigma v \rangle_{DHe^3}$ the velocity average of the fusion cross-section, and W_{ji} the rate of energy transfer to the fuel ions from the species j where $j = f\alpha, fp, e, \alpha, p$ represent fast alpha, fast (fusion) protons which are generated by the D-He³ reaction, electrons, thermalized alphas, and thermalized protons, respectively. Other notations follow the standard definitions in plasma physics.

The conservation equations for the electrons in the plasma can be written as

$$\frac{dn_e}{dt} = S_i - \frac{n_e}{\tau_e} \quad (7)$$

and

$$\begin{aligned} \frac{d(n_e E_e)}{dt} = & S_i E_{in}^e - \frac{n_e}{\tau_e} E_{Le} + W_{f\alpha e} \\ & + W_{fp e} + W_{\alpha e} + W_{pe} - W_{ei} - P_b - P_s. \end{aligned} \quad (8)$$

The injection (source) rate of electrons is the same as that of ions; thus, the term $S_i E_{in}^e$ denotes the electron injection power density, where the quantity E_{in}^e represents electron injection energy. The terms P_b and P_s in Eq. (8) denote the bremsstrahlung and synchrotron radiation losses, respectively. The bremsstrahlung power density extended to include protons can be expressed as

$$\begin{aligned} P_b = & 3.34 \times 10^{-15} [Z_i n_i + n_p + n_{fp} + 2(n_\alpha + n_{f\alpha})] \\ & \times [Z_i^2 n_i + n_p + n_{fp} + 4(n_\alpha + n_{f\alpha})] T_e^{1/2} \\ & [\text{keV cm}^{-3} \text{sec}^{-1}], \end{aligned} \quad (9)$$

where Z_i is the average charge state of the fuel ions and is defined as

$$Z_i = C_D + 2C_{He^3}. \quad (10)$$

The synchrotron radiation can be written in the convenient form [23]

$$P_s = \frac{1.56 \times 10^{-16} n_e T_e}{\beta} \left(n_e T_e + \sum_j n_j T_j \right) \times \left(1 + \frac{T_e}{204} \right) (1 - R_e) \quad [\text{keV} \cdot \text{cm}^{-3} \cdot \text{sec}^{-1}], \quad (11)$$

where R_e is the reflection coefficient of the walls surrounding the plasma and β the well-known plasma beta which is defined by

$$\beta = 1 - \left(\frac{B_p}{B_{p0}} \right)^2, \quad (12)$$

where B_p is the magnetic field in the plasma and B_{p0} the vacuum magnetic field. The index j in Eq. (11) denotes the ion species involved in the reaction. In Eqs. (10) and (11), densities and temperatures are measured in cm^{-3} and keV, respectively.

The alpha particles and protons which are generated by the fusion reaction are classified into two categories:

the fast group represented by the energy range that begins with the birth energy (3.67 MeV for alpha particles and 14.67 MeV for protons) and ends with the thermalization energy, and the thermal group which had thermalized on the background plasma and acquired a temperature not much different from that of the ion or the electrons. We have the following conservation equations: for the thermal alphas,

$$\frac{dn_\alpha}{dt} = -\frac{n_\alpha}{\tau_\alpha} + C_D C_{He^3} n_i^2 \langle \sigma v \rangle_{DHe^3} \quad (13)$$

$$\frac{d(n_\alpha E_\alpha)}{dt} = -\frac{n_\alpha}{\tau_\alpha} E_{L\alpha} + C_D C_{He^3} n_i^2 \langle \sigma v \rangle_{DHe^3} E_\alpha + W_{f\alpha\alpha} + W_{fp\alpha} + W_{p\alpha} - W_{\alpha i} - W_{\alpha e} \quad (14)$$

and for the thermal protons,

$$\frac{dn_p}{dt} = -\frac{n_p}{\tau_p} + C_D C_{He^3} n_i^2 \langle \sigma v \rangle_{DHe^3} \quad (15)$$

$$\frac{d(n_p E_p)}{dt} = -\frac{n_p}{\tau_p} E_{Lp} + C_D C_{He^3} n_i^2 \langle \sigma v \rangle_{DHe^3} E_p + W_{f\alpha p} + W_{fpp} - W_{p\alpha} - W_{pi} - W_{pe}. \quad (16)$$

Then, the density and the average energy of the fast alpha particles in the D-He³ reaction are found as

$$n_{f\alpha} = C_D C_{He^3} n_i^2 \langle \sigma v \rangle_{DHe^3} \tau_{f\alpha} \quad (17)$$

and

$$E_{f\alpha} = \frac{(c_2/c_1)^{2/3}}{c_1 \tau_{f\alpha}} \left\{ \epsilon^2 - \left[\frac{1}{3} \ln \left(\frac{\epsilon^2 - \epsilon + 1}{\epsilon^2 + 2\epsilon + 1} \right) + \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2\epsilon - 1}{\sqrt{3}} \right) \right] \right\}_{E_{th}=E_\alpha}^{E_{\alpha 0}}, \quad (18)$$

where $\tau_{f\alpha}$ is the fast alpha particle confinement time which is known to have the form [24]

$$\tau_{f\alpha} = \frac{2}{3c_1} \ln \left[\frac{1 + \left(\frac{c_1}{c_2} \right) E_{\alpha 0}^{3/2}}{1 + \left(\frac{c_1}{c_2} \right) E_\alpha^{3/2}} \right]. \quad (19)$$

The other notations are defined as

$$\epsilon = \left(\frac{c_1}{c_2} \right)^{1/3} E^{1/2} \quad (20)$$

$$c_1 = \frac{64\sqrt{\pi} n_e e^4 m_e^{1/2}}{3\sqrt{2} m_\alpha T_e^{3/2}} \ln \Lambda_{f\alpha e} \quad (21)$$

$$c_2 = c_i + c_\alpha + c_p \quad (22)$$

$$c_i = \frac{16\sqrt{\pi} Z_i^2 n_i e^4 m_\alpha^{1/2}}{\sqrt{2} m_i} \ln \Lambda_{f\alpha i} \quad (23)$$

$$c_\alpha = \frac{64\pi n_\alpha e^4}{\sqrt{2} m_\alpha^{1/2}} \ln \Lambda_{f\alpha\alpha} \quad (24)$$

$$c_p = \frac{16\pi n_p e^4 m_\alpha^{1/2}}{\sqrt{2} m_p} \ln \Lambda_{f\alpha p}. \quad (25)$$

Likewise, for the fast protons, we have

$$n_{fp} = C_D C_{He^3} n_i^2 \langle \sigma v \rangle_{DHe^3} \tau_{fp} \quad (26)$$

and

$$E_{fp} = \frac{(p_2/p_1)^{2/3}}{p_1 \tau_{fp}} \left\{ \epsilon^2 - \left[\frac{1}{3} \ln \left(\frac{\epsilon^2 - \epsilon + 1}{\epsilon^2 + 2\epsilon + 1} \right) + \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2\epsilon - 1}{\sqrt{3}} \right) \right] \right\}_{E_{th}=E_p}^{E_{p0}}, \quad (27)$$

where the fast proton confinement time and other symbols are given by

$$\tau_{fp} = \frac{2}{3p_1} \ln \left[\frac{1 + \left(\frac{p_1}{p_2}\right) E_{p0}^{3/2}}{1 + \left(\frac{p_1}{p_2}\right) E_p^{3/2}} \right] \quad (28)$$

$$\epsilon = \left(\frac{p_1}{p_2}\right)^{1/3} E^{1/2} \quad (29)$$

$$p_1 = \frac{64\sqrt{\pi}n_e e^4 m_e^{1/2}}{3\sqrt{2}m_p T_e^{3/2}} \ln \Lambda_{fpe} \quad (30)$$

$$p_2 = p_i + p_\alpha + p_p \quad (31)$$

$$p_i = \frac{4\pi Z_i^2 n_i e^4 m_p^{1/2}}{\sqrt{2}m_i} \ln \Lambda_{fpi} \quad (32)$$

$$p_\alpha = \frac{16\pi n_\alpha e^4 m_p^{1/2}}{\sqrt{2}m_\alpha} \ln \Lambda_{fpa} \quad (33)$$

$$p_p = \frac{4\pi n_p e^4}{\sqrt{2}m_p^{1/2}} \ln \Lambda_{fpp}. \quad (34)$$

The various energy transfer rate terms in Eqs. (6), (8), (14) and (16) can be obtained as follows: If the portion of the alpha energy that goes to the electrons is given by

$$Q_e = c_1 \tau_{f\alpha} E_{f\alpha}, \quad (35)$$

then the rate of energy transfer to the electrons can be expressed as

$$W_{f\alpha e} = \frac{n_{f\alpha} Q_e}{\tau_{f\alpha}} = c_1 n_{f\alpha} E_{f\alpha}. \quad (36)$$

When we subtract this quantity from the total energy transfer to all the species, we obtain the portion deposited in the ions, thermal alphas, and thermal protons together, *i.e.*,

$$W_{f\alpha(i+\alpha+p)} = \frac{n_{f\alpha}}{\tau_{f\alpha}} (E_{\alpha 0} - E_\alpha) - W_{f\alpha e}, \quad (37)$$

which upon apportionment yields

$$W_{f\alpha i} = \frac{c_i}{c_2} \left[\frac{n_{f\alpha}}{\tau_{f\alpha}} (E_{\alpha 0} - E_\alpha) - W_{f\alpha e} \right] \quad (38)$$

$$W_{f\alpha \alpha} = \frac{c_\alpha}{c_2} \left[\frac{n_{f\alpha}}{\tau_{f\alpha}} (E_{\alpha 0} - E_\alpha) - W_{f\alpha e} \right] \quad (39)$$

$$W_{f\alpha p} = \frac{c_p}{c_2} \left[\frac{n_{f\alpha}}{\tau_{f\alpha}} (E_{\alpha 0} - E_\alpha) - W_{f\alpha e} \right]. \quad (40)$$

In a similar manner, we can show the rates of energy transfer to the thermal species in the plasma from the fast protons:

$$W_{fpe} = p_1 n_{fp} E_{fp} \quad (41)$$

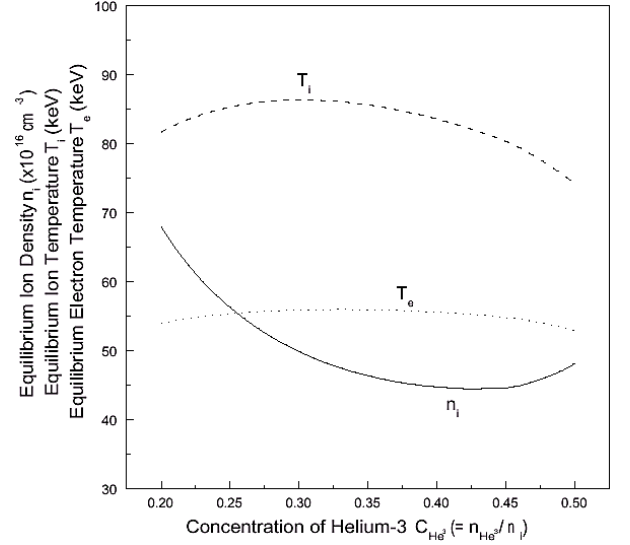


Fig. 1. Variations of density and temperatures with the concentration of helium-3 ($Q = 1$, $\beta = 0.95$, $R = 50$, $L = 50$ m, $E_{in} = 200$ keV).

$$W_{fpi} = \frac{p_i}{p_2} \left[\frac{n_{fp}}{\tau_{fp}} (E_{p0} - E_p) - W_{fpe} \right] \quad (42)$$

$$W_{fpa} = \frac{p_\alpha}{p_2} \left[\frac{n_{fp}}{\tau_{fp}} (E_{p0} - E_p) - W_{fpe} \right] \quad (43)$$

$$W_{fpp} = \frac{p_p}{p_2} \left[\frac{n_{fp}}{\tau_{fp}} (E_{p0} - E_p) - W_{fpe} \right]. \quad (44)$$

As for the rate of energy exchange between the thermal species of kind j and k , the standard formula can be readily applied, namely [25],

$$W_{jk} = \frac{n_j n_k}{(n\tau)_{jk}} (E_j - E_k) \quad (45)$$

$$(n\tau)_{jk} = \frac{\sqrt{2}m_j m_k}{8\sqrt{\pi}Z_j^2 Z_k^2 e^4} \left(\frac{T_j}{m_j} + \frac{T_k}{m_k} \right)^{3/2} \frac{1}{\ln \Lambda_{jk}} \quad (46)$$

where Z_j and Z_k denote the charge numbers of the interacting particles j and k .

III. RESULTS

The conservation equations for D-He³, which are Eqs. (5) to (8) and (13) to (16), are solved numerically, and the results are shown in Figs. 1 to 6. In the numerical calculation, the plasma beta and the mirror ratio are fixed at 0.95 and 50, respectively. For Figs. 1 to 4, the value of the plasma length (mirror to mirror) is 50 m. To decide the best combination of D and He³ in the gas-dynamic mirror device for a steady-state operation, we examined the variations of the plasma and propulsion

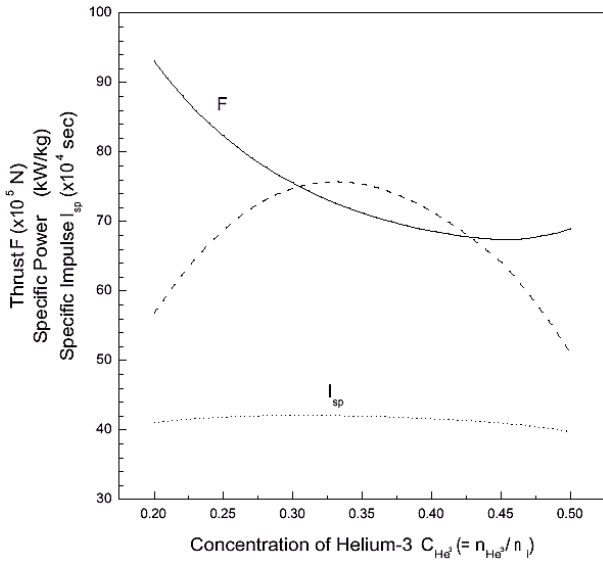


Fig. 2. Variations of thrust, specific power and specific impulse with the concentration of helium-3 ($Q = 1$, $\beta = 0.95$, $R = 50$, $L = 50$ m, $E_{in} = 200$ keV).

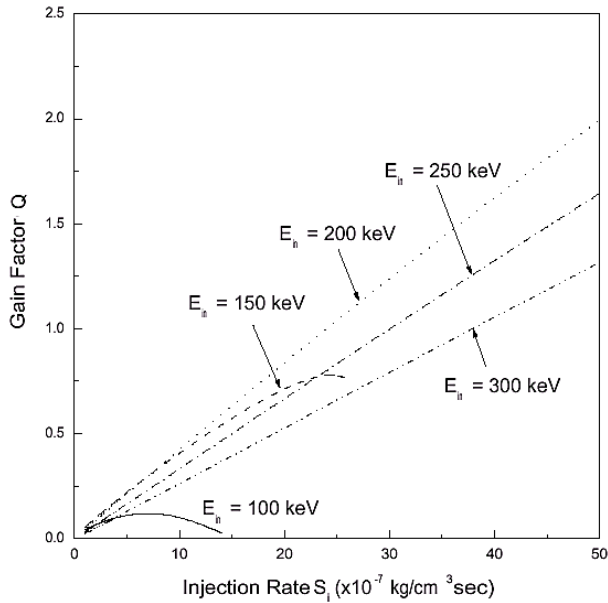


Fig. 3. Role of injection energy in the gain factor ($\beta = 0.95$, $R = 50$, $L = 50$ m).

parameters with the concentration of He^3 , *i.e.*, C_{He^3} , in the fuel. As can be seen in Fig. 1, the ion density has a minimum when the concentration of He^3 is 0.43 (= 43 %) while the ion and the electron temperatures have maximum values when the concentrations are 0.3 (= 30 %) and 0.33 (= 33 %), respectively. The ion and the electron temperatures are sensitive to the specific impulse, which is one of the most important figures of merit for space propulsion. The specific impulse is the effective exhaust velocity divided by the gravitational acceleration

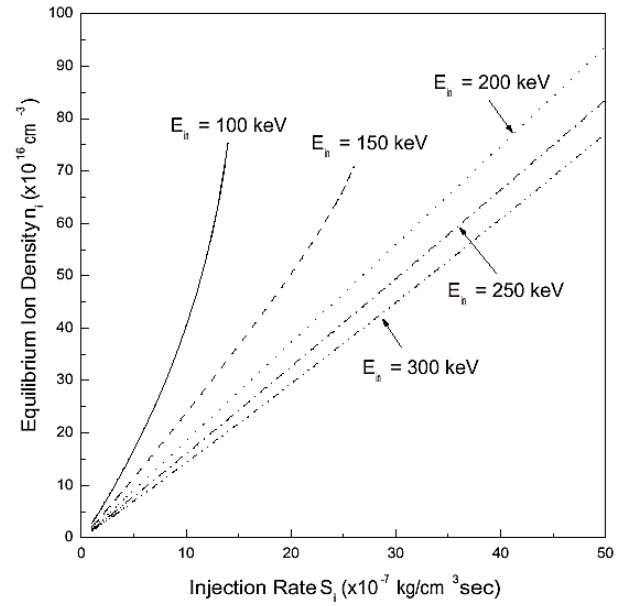


Fig. 4. Variation of the ion (D-He^3) density with injection rate for several injection energies ($\beta = 0.95$, $R = 50$, $L = 50$ m).

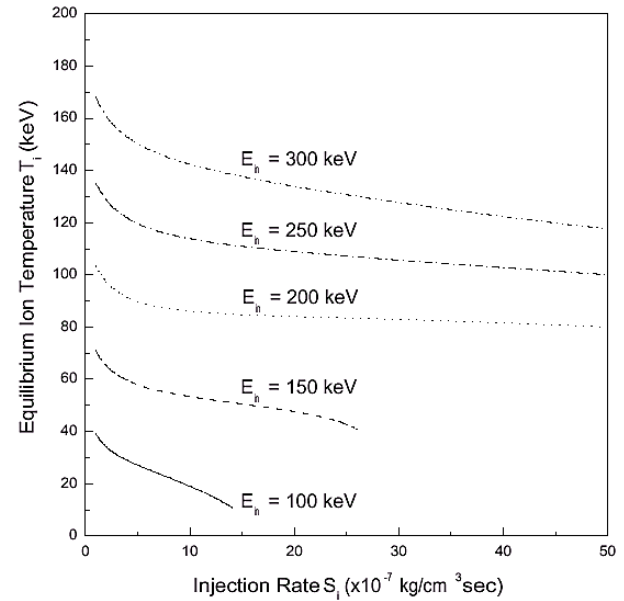


Fig. 5. Variation of the ion (D-He^3) temperature with injection rate for several injection energies ($\beta = 0.95$, $R = 50$, $L = 50$ m).

constant. Therefore, a higher specific impulse can give a higher thrust and significantly reduce the space trip time. As shown in Fig. 2, obviously one can see that the specific impulse has a similar pattern as the temperatures with its maximum of 4.21×10^5 seconds at $C_{\text{He}^3} = 0.31$. From the propulsion point of view, high temperature and high density could be desirable since they would yield high thrust. However, if the density increases

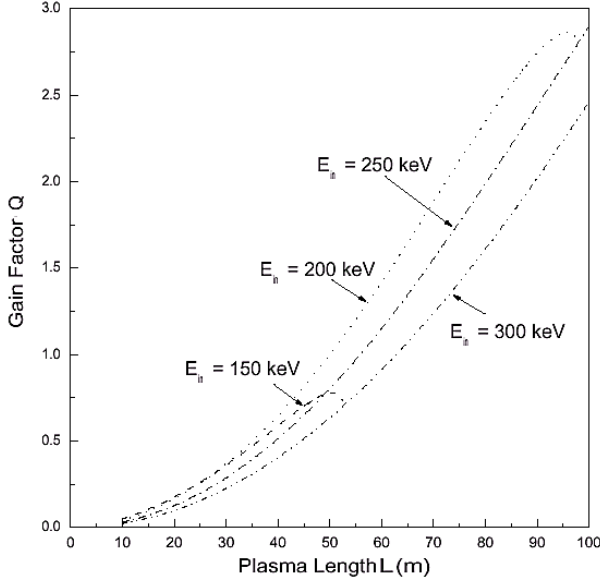


Fig. 6. Variation of the gain factor with plasma length for several injection energies ($\beta = 0.95$, $R = 50$, $S_i = 2.4 \times 10^{-6}$ kg/cm³-sec).

sharply, as shown in the range of $C_{He^3} = 0.2 \sim 0.3$ in Fig. 1, it causes high bremsstrahlung and synchrotron radiation powers, which will result in an undesirable rocket mass increase. This fact is well shown in the variation of specific power (*i.e.*, the ratio of thrust power to the total mass) in Fig. 2. One can see that, although the thrust or thrust power is increasing as the concentration of He³ is decreasing, the specific power is decreasing due to the mass increase caused by the density increase. The specific power has a maximum value of 75.73 kW/kg at $C_{He^3} = 0.33$. By considering all these factors, we found that values of C_{He^3} between 0.3 and 0.4 seem moderately good.

The variation of the gain factor Q with the fuel ion (D-He³) injection rate S_i for different injection energies E_{in} is displayed in Fig. 3. We observe that the break-even condition, $Q = 1$, cannot be achieved for injection energies equal to or less than 150 keV. The figure shows that when the injection energy is equal to or higher than 200 keV, the gain factor increases beyond unity as the fuel ion injection rate increases; however, at a fixed S_i , a lower gain factor is achieved for injection energies higher than 200 keV. Figures 4 and 5 show the variations of the steady-state ion (D-He³) density and temperature with the fuel ion injection rate. The role of the injection energy with a fixed injection rate in the variation of the gain factor with plasma length is depicted in Fig. 6. For example, to obtain the break-even condition, $Q = 1$, at a plasma length $L = 50$ m, a plasma beta $\beta = 0.95$, a mirror ratio $R = 50$, and an injection energy $S_i = 2.4 \times 10^{-6}$ kg/cm³-sec, we must have an injection energy of 200 keV. If we increase the injection energy, say, $E_{in} = 250$ keV, then an increase in the plasma length to L

Table 1. Effect of helium-3 concentration on the steady-state operating parameters of a D-He³ GDM system ($Q = 1$, $\beta = 0.95$, $R = 50$, $L = 50$ m, $E_{in} = 200$ keV).

Parameter	Helium-3 Concentration C_{He^3}		
	0.2	0.3	0.4
Ion Density n_i (cm ⁻³)	6.8×10^{17}	5.0×10^{17}	4.5×10^{17}
Ion Temperature T_i (keV)	81.7	86.4	83.6
Electron Temperature T_e (keV)	54.0	55.9	55.6
Fusion Power P_f (MW)	2.2×10^7	1.7×10^7	1.5×10^7
Neutron Power P_n (MW)	5.1×10^5	1.3×10^5	2.5×10^4
Thrust F (N)	9.3×10^6	7.6×10^6	6.9×10^6
Specific Impulse I_{sp} (sec)	4.1×10^5	4.2×10^5	4.2×10^5

$= 56$ m is required to maintain the same steady-state operating conditions. These results can be adopted to design the optimized D-He³ fueled propulsion device.

Some numerical values for the D-He³ GDM system are also shown in Table 1. With reasonable engineering and technical assumptions, although they are not explicitly explained here, the two most important propulsion parameters, namely, the specific impulse and the thrust, were calculated. Detailed works and calculations can be seen in the Refs. 20 and 21. Because of the high temperatures of a D-He³ GDM system, it was found in this work that the specific impulse was increased by almost a factor of four relative to the D-T case shown in Ref. 20, while the thrust was increased by a factor of about 136 due to the higher density.

IV. CONCLUSIONS

Because of the radioactivity that can be induced by the neutrons in some typical fusion reactions, D-He³ is the most often mentioned fuel due to its minimal production of neutrons and its reaction products being charged particles. The charged particles produced could be used effectively in an energy conversion scheme that directly converts their energy into electrical power that could be used to drive the system [26]. When applied to the GDM system, a D-He³ fuel is seen to require a very high injection energy (≥ 200 keV) to produce $Q > 1$ in a steady-state operation. Obviously enough, the fact that the propulsion characteristics of the D-He³ case far exceed its counterpart in the D-T case will lead to a very sizable and massive system in practice because higher bremsstrahlung and synchrotron radiations will require additional radiators and so on. However, the redeeming factor for the advanced fuel case, nevertheless, is that it will reduce the space travel time significantly because the travel time depends more on the specific impulse than it does on the mass of the system, as well known. These high performance characteristics will, indeed, make their presence felt in the crewed interplanetary missions of the future.

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REFERENCES

- [1] R. B. Adams, G. Statham, S. White, B. Patton, Y. C. F. Thio, R. Alexander, S. Fincher, T. Polsgrove, J. Chapman, R. Hopkins and A. Philips, *39th AIAA/ASME/SAE/ASEE Joint Propulsion Conference and Exhibit* (Huntsville, AL, July 20-23, 2003), AIAA 2003-4527.
- [2] S. J. Hoffman, NASA Technical Paper TP-2001-209371, 2001.
- [3] F. R. Chang-Diaz, *Scientific American* (Nov., 2000), p. 90.
- [4] T. Kammash, *50th International Astronautical Congress* (Amsterdam, Netherlands, 1999), IAF-99-S.6.01.
- [5] S. J. Hoffman and D. A. Kaplan, *NASA Special Publication* (Houston, 1997), SP-6107.
- [6] J. L. Bromberg, *Fusion* (MIT Press, Cambridge, 1982).
- [7] J.-S. Ko, D.-K. Kim, S. H. Hong and K.-H. Im, *J. Korean Phys. Soc.* **41**, 212 (2002).
- [8] O. Kwon, *J. Korean Phys. Soc.* **42**, 118 (2003).
- [9] E. Teller, A. J. Glass, T. K. Fowler, A. Hasegawa and J. F. Santarius, *Fusion Tech.* **22**, 82 (1992).
- [10] R. F. Post and J. F. Santarius, *Fusion Tech.* **22**, 13 (1992).
- [11] J. F. Santarius and B. G. Logan, *J. Propulsion and Power* **14**, 519 (1998).
- [12] A. R. Martin and A. Bond, *J. British Interplanetary Soc. Supplement* **31**, S44 (1978); A. Bond and A. R. Martin, *J. British Interplanetary Soc. Supplement* **31**, S63 (1978).
- [13] C. D. Orth, *12th Symposium on Fusion Engineering* (IEEE, New York, 1987), p. 1017.
- [14] T. Kammash and D. L. Galbraith, *Fusion Tech.* **12**, 11 (1987).
- [15] T. Kammash and D. L. Galbraith, *J. Propulsion and Power* **6**, 412 (1990).
- [16] R. B. Adams, R. Alexander, J. Chapman, S. Fincher, R. Hopkins, A. Philips, T. Polsgrove, R. Litchford, B. Patton, G. Statham, S. White and Y. C. F. Thio, *Conceptual Design of in-Space Vehicles for Human Exploration of the Outer Planets* (Nov., 2003), NASA-TP-2003-212691.
- [17] G. Statham, S. White, R. B. Adams, Y. C. F. Thio, R. Alexander, S. Fincher, A. Philips and T. Polsgrove, *39th AIAA/ASME/SAE/ASEE Joint Propulsion Conference and Exhibit* (Huntsville, AL, July 20-23, 2003), AIAA 2003-4526.
- [18] B. N. Cassenti, T. Kammash and D. L. Galbraith, *J. Propulsion and Power* **13**, 428 (1997).
- [19] T. Kammash, *Antimatter Driven p -B¹¹ Fusion Propulsion System, Space Technology and Applications International Forum* (Albuquerque, NM, Feb. 2-5, 2003).
- [20] T. Kammash and M.-J. Lee, *J. Propulsion and Power* **11**, 3 (1995).
- [21] T. Kammash, M.-J. Lee and D. I. Poston, *J. Propulsion and Power* **13**, 421 (1997).
- [22] T. Kammash and D. L. Galbraith, *J. Propulsion and Power* **14**, 24 (1998).
- [23] D. P. Kilcrease and R. C. Kirkpatrick, *Nuclear Fusion* **28**, 1465 (1988).
- [24] T. Kammash, *J. Propulsion and Power* **16**, 1100 (2000).
- [25] T. Kammash, *Fusion Reactor Physics, Principles and Technology* (Ann Arbor Science Publishers, Ann Arbor, 1975), Chap. 4.
- [26] T. Kammash and M.-J. Lee, *32nd AIAA/ASME/SAE/ASEE Joint Propulsion Conference and Exhibit* (Lake Buena Vista, FL, July 1-3, 1996), AIAA 96-3066.